GraphSmile: paper tables and reproduced results

Tracing intricate cues in dialogue (IEEE TPAMI, 2025). Paper values against our re-run.

The paper contains seven tables, Tables I to VII. Each table below follows the paper's layout. Rows marked **paper** are the paper's numbers. Rows marked **ours** are our reproduced numbers, from the logs in *results/logs/*. **Δ** is ours minus paper.

- Table I: Merging scheme of emotion labels
- Table II: Statistical information of all datasets
- Table III: Description of partial hyperparameters
- Table IV: Main results for the MERC task on all datasets
- Table V: Main results for the MSAC task on all datasets
- Table VI: Results for ablation studies on all datasets
- Table VII: Time and memory overhead for the MERC task on all datasets

TABLE I

MERGING SCHEME OF EMOTION LABELS

Maps each dataset's emotion labels to negative, neutral and positive sentiment, so that every dataset shares one sentiment space.

Sentiment	IEMOCAP-6	IEMOCAP-4	MELD	CMU-MOSEI
Negative	Sad, Angry, Frustrated	Sad, Angry	Negative	Highly Negative, Negative, Weakly Negative
Neutral	Neutral	Neutral	Neutral	Neutral
Positive	Happy, Excited	Happy	Positive	Weakly Positive, Positive, Highly Positive

Source: paper, Table I.

TABLE II

STATISTICAL INFORMATION OF ALL DATASETS

Number of conversations and utterances in each dataset, split into train+validation and test.

Dataset	Conversation train+val	Conversation test	Utterance train+val	Utterance test
IEMOCAP-6	120	31	5,810	1,623
IEMOCAP-4	120	31	3,600	943
MELD	1,153	280	11,098	2,610
CMU-MOSEI	2,549	646	18,198	4,662

The 10% of dialogues from the training and validation sets are used as the validation set. Source: paper, Table II. Not reproduced: this table describes the data.

TABLE III

DESCRIPTION OF PARTIAL HYPERPARAMETERS

Per-dataset learning rate, batch size, dropout, layer count, window sizes and loss weights used in the main experiments.

Dataset	LR	BS	DR	L	P	B	λs	λo
IEMOCAP-6	1e-4	16	0.2	7	17	19	1.0	0.7
IEMOCAP-4	3e-4	16	0.2	4	5	10	0.6	0.6
MELD	7e-5	16	0.2	5	3	3	0.5	0.2
CMU-MOSEI	8e-5	32	0.4	2	5	2	0.8	1.0

LR, BS and DR are the learning rate, batch size and dropout rate. L is the number of heterogeneous layers, P and B are the past window and the shift window, and λs and λo are the loss weights. The L2 regularization factor is 1e-3 for all datasets (paper, Section V). Our final runs use these values, including L2 = 1e-3. Source: paper, Table III.

TABLE IV
MAIN RESULTS FOR THE MERC TASK ON ALL DATASETS

Emotion recognition (MERC) results for GraphSmile, per emotion class with ACC and weighted F1 (%).

IEMOCAP-6

Method	Happy	Sad	Neutral	Angry	Excited	Frustrated	ACC	WF1
GraphSmile (paper)	63.09	83.16	71.07	71.38	79.66	66.84	72.77	72.81
GraphSmile (ours)	**54.87**	**82.14**	**70.85**	**68.92**	**78.04**	**66.22**	**71.23**	**71.17**
Δ	*-8.22*	*-1.02*	*-0.22*	*-2.46*	*-1.62*	*-0.62*	*-1.54*	*-1.64*

IEMOCAP-4

Method	Happy	Sad	Neutral	Angry	ACC	WF1
GraphSmile (paper)	87.94	88.67	85.41	84.73	86.53	86.52
GraphSmile (ours)	**86.12**	**88.45**	**84.70**	**85.22**	**86.00**	**85.98**
Δ	*-1.82*	*-0.22*	*-0.71*	*+0.49*	*-0.53*	*-0.54*

MELD

Method	Neutral	Surprise	Fear	Sadness	Joy	Disgust	Anger	ACC	WF1
GraphSmile (paper)	80.35	59.11	18.18	42.46	64.99	32.43	53.67	67.70	66.71
GraphSmile (ours)	**80.27**	**58.67**	**19.18**	**43.45**	**64.56**	**29.36**	**53.46**	**67.59**	**66.55**
Δ	*-0.08*	*-0.44*	*+1.00*	*+0.99*	*-0.43*	*-3.07*	*-0.21*	*-0.11*	*-0.16*

CMU-MOSEI

Method	HNeg	Neg	WNeg	Neu	WPos	Pos	HPos	ACC	WF1
GraphSmile (paper)	0.00	28.79	43.50	39.19	57.30	35.95	6.25	46.82	44.93
GraphSmile (ours)	**0.00**	**30.28**	**45.19**	**42.81**	**53.40**	**33.39**	**3.17**	**45.10**	**43.96**
Δ	*+0.00*	*+1.49*	*+1.69*	*+3.62*	*-3.90*	*-2.56*	*-3.08*	*-1.72*	*-0.97*

The paper also lists baseline rows (COGMEN, Joyful, DER-GCN, DCGCN, MMGCN, MM-DFN, M3Net, SACL-LSTM, DialogueCRN, MetaDrop, UniMSE, EmoLR). These were not re-run here, so only the GraphSmile rows are shown. Ours: best-test-F1 epoch, Table III settings. Δ = ours minus paper. Source: paper, Table IV; reproduced values from results/comparison.md.

TABLE V
MAIN RESULTS FOR THE MSAC TASK ON ALL DATASETS

Sentiment (MSAC) results for GraphSmile. † uses the Table III settings; ‡ uses settings optimized for MSAC.

IEMOCAP-6

Method	Negative	Neutral	Positive	ACC	WF1
GraphSmile† (paper)	86.87	69.06	89.21	83.73	83.66
GraphSmile† (ours)	**87.11**	**70.26**	**89.66**	**83.67**	**83.82**
Δ	*+0.24*	*+1.20*	*+0.45*	*-0.06*	*+0.16*
GraphSmile‡ (paper)	88.28	72.89	89.93	84.97	85.09

IEMOCAP-4

Method	Negative	Neutral	Positive	ACC	WF1
GraphSmile† (paper)	87.76	82.93	87.23	85.79	85.71
GraphSmile† (ours)	**87.73**	**82.21**	**85.71**	**85.26**	**85.18**
Δ	*-0.03*	*-0.72*	*-1.52*	*-0.53*	*-0.53*
GraphSmile‡ (paper)	89.45	85.53	87.02	87.49	87.48

MELD

Method	Neutral	Positive	Negative	ACC	WF1
GraphSmile† (paper)	79.15	64.84	70.08	73.60	73.40
GraphSmile† (ours)	**79.09**	**64.18**	**70.07**	**73.45**	**73.24**
Δ	*-0.06*	*-0.66*	*-0.01*	*-0.15*	*-0.16*
GraphSmile‡ (paper)	79.35	67.89	70.72	74.44	74.31

CMU-MOSEI

Method	Negative	Neutral	Positive	ACC	WF1
GraphSmile† (paper)	56.14	39.80	78.18	66.83	65.69
GraphSmile† (ours)	**58.29**	**42.10**	**76.65**	**65.27**	**65.61**
Δ	*+2.15*	*+2.30*	*-1.53*	*-1.56*	*-0.08*
GraphSmile‡ (paper)	56.99	41.02	79.36	67.73	66.73

Ours (†) is reproduced from the sentiment head at the best-test-F1 epoch. The ‡ (optimized) rows are the paper's values only, because the paper does not give the optimized settings, so they are not reproduced. Source: paper, Table V; reproduced values from results/comparison.md.

TABLE VI
RESULTS FOR ABLATION STUDIES ON ALL DATASETS

Ablations for the MERC task (top) and the MSAC task (bottom): each variant with the paper's value and ours.

MERC task (emotion)

		IEMOCAP-6		IEMOCAP-4		MELD		CMU-MOSEI	
Variant	**Source**	**ACC**	**WF1**	**ACC**	**WF1**	**ACC**	**WF1**	**ACC**	**WF1**
GraphSmile	paper	72.77	72.81	86.53	86.52	67.70	66.71	46.82	44.93
	ours	**71.23**	**71.17**	**86.00**	**85.98**	**67.59**	**66.55**	**45.10**	**43.96**
No L_s	paper	71.53	71.52	85.15	85.15	67.39	66.30	45.08	43.37
	ours	**70.92**	**70.83**	**85.37**	**85.36**	**67.43**	**66.33**	**45.43**	**44.43**
No L_o	paper	71.41	71.42	84.52	84.50	67.62	66.59	45.20	44.03
	ours	**70.92**	**71.02**	**84.84**	**84.79**	**67.59**	**66.58**	**45.53**	**44.09**
No L_s & L_o	paper	70.67	70.49	84.41	84.40	67.55	66.43	44.29	43.54
	ours	**70.06**	**70.01**	**85.05**	**85.04**	**67.51**	**66.40**	**44.74**	**43.86**
No Res	paper	61.98	61.35	79.64	79.58	52.07	46.24	41.75	40.12
	ours	**61.37**	**61.34**	**80.28**	**80.21**	**48.24**	**47.36**	**43.02**	**40.25**
No FC-Res	paper	70.43	70.39	85.37	85.36	67.36	66.30	45.03	43.28
	ours	**70.43**	**70.47**	**85.26**	**85.25**	**67.39**	**66.35**	**45.12**	**43.41**
No Seg	paper	71.66	71.69	85.47	85.43	67.59	66.59	45.89	44.53
	ours	**71.41**	**71.32**	**85.05**	**85.01**	**67.66**	**66.64**	**45.67**	**44.22**
No Textual	paper	55.70	54.23	62.88	62.22	50.15	42.27	32.98	28.28
	ours	**59.33**	**59.34**	**64.16**	**62.92**	**50.11**	**43.55**	**35.54**	**31.19**
No Visual	paper	71.29	71.23	86.21	86.20	67.39	66.26	45.32	43.57
	ours	**70.98**	**71.06**	**85.79**	**85.76**	**67.47**	**66.45**	**45.65**	**43.95**
No Acoustic	paper	68.45	68.26	83.35	83.28	67.55	66.40	46.56	44.57
	ours	**67.59**	**67.52**	**83.14**	**83.14**	**67.36**	**66.27**	**45.89**	**44.08**

MSAC task (sentiment)

		IEMOCAP-6		IEMOCAP-4		MELD		CMU-MOSEI	
Variant	**Source**	**ACC**	**WF1**	**ACC**	**WF1**	**ACC**	**WF1**	**ACC**	**WF1**
GraphSmile	paper	84.97	85.09	87.49	87.48	74.44	74.31	67.73	66.73
	ours	**83.67**	**83.82**	**85.26**	**85.18**	**73.45**	**73.24**	**65.27**	**65.61**
No L_e	paper	84.41	84.53	85.47	85.40	73.98	73.84	66.59	65.91
	ours	**54.59**	**42.74**	**82.93**	**82.75**	**72.61**	**72.33**	**65.58**	**65.39**
No L_o	paper	84.10	84.18	85.26	85.23	74.18	74.10	66.71	66.20
	ours	**82.26**	**82.45**	**83.46**	**83.25**	**73.52**	**73.32**	**66.73**	**65.03**
No L_e & L_o	paper	83.30	83.54	85.58	85.55	73.91	73.77	66.54	65.64
	ours	**80.71**	**80.20**	**44.01**	**26.90**	**72.72**	**72.45**	**58.58**	**59.98**
No Res	paper	78.56	78.25	81.02	80.83	58.08	56.61	62.26	61.06
	ours	**76.96**	**76.02**	**81.34**	**81.13**	**55.67**	**55.60**	**62.98**	**61.50**
No FC-Res	paper	84.35	84.38	86.00	85.90	73.72	73.64	66.20	65.69
	ours	**83.43**	**83.72**	**85.15**	**85.02**	**73.33**	**73.12**	**66.37**	**64.54**
No Seg	paper	83.49	83.64	85.37	85.32	74.37	74.22	66.44	65.88
	ours	**83.24**	**83.46**	**84.31**	**84.14**	**73.41**	**73.20**	**66.85**	**65.87**
No Textual	paper	71.04	70.42	66.28	60.76	52.72	52.10	54.95	47.80
	ours	**71.16**	**70.97**	**64.48**	**62.31**	**51.72**	**50.62**	**55.04**	**48.93**
No Visual	paper	84.53	84.69	85.58	85.57	73.83	73.75	66.30	65.81
	ours	**84.35**	**84.32**	**85.37**	**85.27**	**73.26**	**73.03**	**66.20**	**64.80**
No Acoustic	paper	83.24	82.87	84.31	84.25	73.72	73.62	66.47	65.57
	ours	**82.32**	**82.05**	**84.31**	**84.23**	**73.22**	**72.97**	**66.80**	**65.34**

All Table VI rows are run. The MSAC block in the paper appears to use the optimized sentiment settings (‡), which we have not confirmed for every row; ours use Table III (†). No L_s and No L_s & L_o remove the sentiment loss, so their MSAC cells are not comparable. No L_e and No L_e & L_o are far below the paper on IEMOCAP-6 and IEMOCAP-4 (for example 54.59 against 84.41 ACC on IEMOCAP-6, and 44.01 against 85.58 on IEMOCAP-4); these runs are not verified and may not have converged. Source: paper, Table VI; reproduced values from results/comparison.md.

TABLE VII
TIME AND MEMORY OVERHEAD FOR THE MERC TASK ON ALL DATASETS

Seconds per epoch and allocated memory for GraphSmile, paper against ours.

Dataset	Paper: time (s/epoch)	Ours: time (s/epoch)	Paper: memory (MB)	Ours: memory (MB)
IEMOCAP-6	1.12	1.61	1840.89	1731.51
IEMOCAP-4	0.67	1.01	509.88	431.89
MELD	5.21	7.55	513.33	422.61
CMU-MOSEI	7.38	12.84	467.13	306.57

Time includes training and testing for one epoch. The paper used an NVIDIA RTX 3090; our runs used a shared NVIDIA A100-PCIE-40GB, so time is not directly comparable. Memory is measured by the code at epoch 2. Source: paper, Table VII; reproduced values from results/comparison.md.